Evaluating LLM Uncertainty Estimation Methods for Real-World Task Routing

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Abstract

As large language models (LLMs) become increasingly integrated into more and more domains, effective human-AI collaboration becomes essential due to the many real-world tasks required to balance between automated efficiency and human agents oversight, making it important for large language models to recognize when they are likely to be incorrect. This study evaluates three uncertainty estimation methods: Verbalized Confidence Elicitation (VCE), Maximum Sequence Probability (MSP), and Sample Consistency (SC) in a realistic setting involving the grading of medical students' clinical notes. Using a benchmark on real-world data with varying levels of ambiguity and difficulties, we compare these methods based on raw benchmark accuracy as well as confidence calibration and discrimination ability. The results show that VCE significantly outperforms the other methods on both confidence metrics, suggesting that directly prompting models to assess their own confidence could potentially provide both better performance and interpretability. However, none of the methods had sufficient performance to justify real-world use, highlighting the necessity of further research and development on the road to effective human-AI collaboration.

1. Introduction & Motivation

As large language models become increasingly integrated into real-world applications, especially in high-stakes domains such as healthcare, finance, and education, the need for effective human-AI collaboration has become critical. Even if LLMs continue to rapidly improve in capability and intelligence, "numerous use cases inherently require agents to collaborate with humans due to humans' latent preferences, domain expertise, or the need for control" (Shao et al., 2024). Instead, the ideal system would have to combine the efficiency of AI with the reliability of human intervention, where the model handles routine tasks and defers uncertain or complex cases to a human agent.

A central challenge in enabling this collaboration is determining when a task requires this human intervention. As this implicitly requires a good measure of the probability that the LLM does not complete the task correctly, a promising direction is to utilize an estimation of the model's confidence or uncertainty. Note here that if the estimation is overly confident in incorrect outputs, it may fail to escalate critical cases. Conversely, if it is too conservative, it may overwhelm human reviewers with unnecessary escalations. Neither situation is ideal, which stresses the importance of achieving an accurate assessment of model confidence.

Despite its importance, uncertainty estimation in large language models (LLMs) remains an open research problem. Prior work has explored several methods, but most evaluations are conducted on benchmark datasets such as question-answering tasks, which do not fully capture the complexity of real-world decision-making scenarios (Hobelsberger et al., 2025).

This study will address this gap by evaluating three widely used uncertainty estimation methods in a realistic setting: grading student-written clinical notes using a structured rubric. The goal is to determine which method best supports practical task routing by aligning model confidence with correctness and ambiguity.

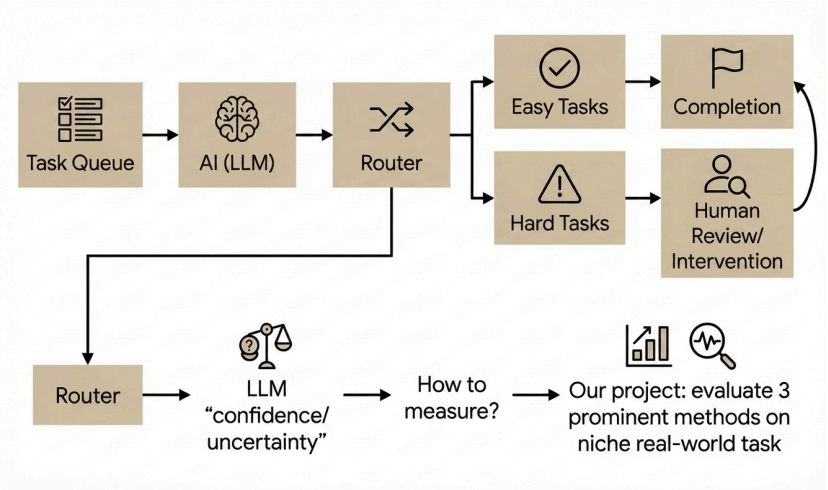

Figure 1. *Diagram succinctly summarizing this section.*

2. Methodology

The dataset used in this study is created by researchers from Rutgers Robert Wood Johnson Medical School and consists of 30 clinical notes written by medical students. Each note is evaluated using a faculty-designed rubric containing 24 binary criteria, resulting in a total of 720 individual grading items. In addition to the rubric, the dataset includes individual evaluations from three expert faculty members as well as a consensus-based ground truth established through discussion. The benchmark created from this dataset is simply a comparison of LLM performance against this ground truth.

One of the key strengths of this benchmark is the variability in task difficulty. Some rubric items are straightforward, such as checking whether a patient's age is presented in the clinical notes, while others are inherently ambiguous and subject to disagreement among experts. This variation provides an ideal testing environment for uncertainty estimation, as it reflects the conditions that AI systems are likely to encounter in real-world applications (Liu et al., 2025).

In our experimentation, we evaluate and compare the three most basic and widely used LLM uncertainty estimation methods, based on prior research. Many other methods in the literature are either modifications or combinations of one or more of these foundational methods (Hobelsberger et al., 2025). Thus, we reasoned that comparing these on our new benchmark would represent a logical first step that is within the scope of an undergraduate course final project.

The first method, Verbalized Confidence Elicitation (VCE), involves directly prompting the model to provide a confidence score along with its prediction, representing the model's estimated probability that its answer is correct. In our implementation, we prompted the LLM to output confidence estimations for each grade item within the interval [0.5, 1]. Note that this interval was derived from framing this

estimation as a direct estimation for the probability that the model would get this item correct - due to the binary nature of grades, the minimum of such probability is 0.5. In VCE only, we additionally prompted the LLM to output explicit verbal “reasoning” for its confidence values - this is a key interpretability advantage of this method that is possible due to the confidence value and its rationale coming from the same source, which is the model itself.

The second method, Maximum Sequence Probability (MSP), estimates confidence using the model’s internal token-level probabilities. Specifically, the negative log-likelihood of a generated output is computed by summing the individual probabilities of each token in the sequence, which are obtained through API logprob decoding. Then, in order to obtain a confidence value that is comparable across tasks, this likelihood is normalized into the interval [0,1] and inverted to reflect “confidence” rather than “uncertainty”. Three sharp limitations for this method must be acknowledged here. Firstly, the generated value may not actually be the “maximum” sequence probability, as a temperature of 0.7 is used and thus the tokens with the highest logprobs are not always the ones being used in both the output and confidence calculation. Secondly, utilizing only one token logprob is a quite naive way to measure uncertainty, as it is unable to distinguish between different token probability distributions - we suggest that measuring the entropy of each individual distribution may be a better measure. However, we did not address either of these first two limitations in our implementation in order to stay aligned with the core intent of our study, which is to compare “widely-used” estimation methods as-is (Hobelsberger et al., 2025). The last limitation is that this method cannot properly handle cases where multiple tasks are completed simultaneously in the same output sequence - in the case of our specific benchmark, it is designed such that 3-5 items are to be completed at a time. Thus, we were forced to assign the same MSP to all items that were graded in the same single output. We purposefully kept this flaw in implementation as it represents a key limitation of this method in terms of real-world use, with the evaluation of this real-world usefulness being an essential component of our study’s intent.

The third method, Sample Consistency, involves generating multiple responses for the same input and measuring the degree of agreement among them, using this consistency as a proxy for confidence. In our implementation, we generated N=5 samples for each grading item, and took the majority agreement proportion as the confidence. Thus, the possible confidence values were 0.6, 0.8, or 1.0. For the final grade assigned to the item, we utilized a majority vote between the 5 samples.

Note that although the possible values differ between these three methods, they are semantically equivalent in that a value of 1 represents full 100% confidence in the item, and lower values represent lower confidence.

For the LLM, we used OpenAI’s best open-source model GPT-OSS-120B served via the Fireworks API. It is perhaps one of the cheapest and fastest models on the market, with all of our work totaling to only $0.59 in actual cost and some providers able to achieve speeds of over 700 tokens per second. In addition, even while being cheap and fast, it is still a decent size model with acceptable performance even by current standards, only being a step or two behind frontier capabilities. Importantly, it excels at instruction following and output formatting, which were essential to a model being able to work in our benchmarking and evaluation system. Lastly, its open-source status allows us to access token log probabilities through the API, which was a requirement for us to implement the MSP method.

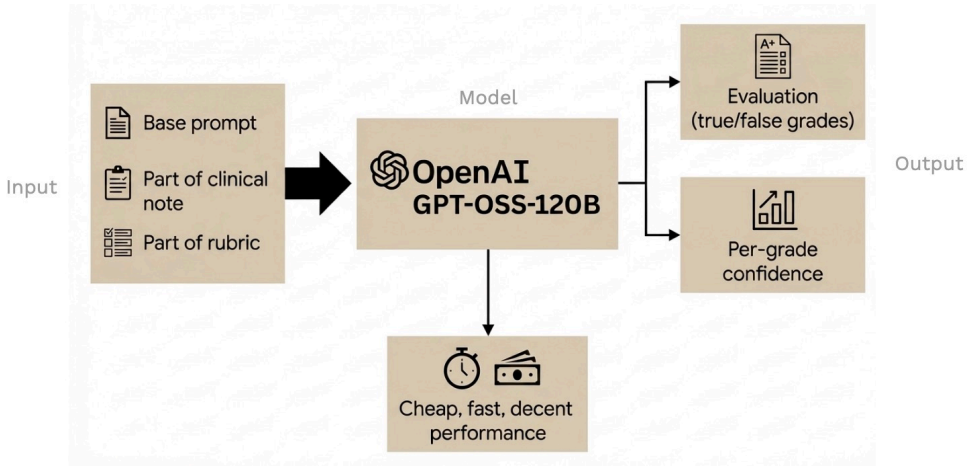

Figure 2. *Overall system architecture.*

To evaluate and compare the three confidence methods, we use three key metrics. Accuracy measures performance on the benchmark while using each method, calculated as the proportion of correct grade items compared to the ground truth evaluations. The interval for accuracy is [0, 1], with higher being better. Expected Calibration Error (ECE) assesses how well the method's confidence aligns with actual accuracy, calculated by splitting the grades into three bins by confidence values, measuring calibration between confidence and actual accuracy for each bin, and taking the weighted average. The interval for ECE is also [0, 1], but lower value (error) is more desirable. Area Under the ROC Curve (AUROC) measures the method's ability to distinguish between correct and incorrect items, calculated as the probability that a randomly chosen correct item received a higher confidence value than a randomly chosen incorrect item. Here the interval is [0, 1] with 0.5 representing no better than random chance and higher being better. Together, these metrics provide a comprehensive view of both predictive performance and the quality of uncertainty estimation (Hobelsberger et al., 2025). Note that only ECE and AUROC specifically measure the quality of confidence estimation values.

3. Results

Metric	Verbalized Confidence	Sequence Probability	Sample Consistency
Accuracy	0.8875	0.8764	0.8778
ECE	**0.0544**	0.3140	0.0939
AUROC	**0.7642**	0.6096	0.6616

Figure 3*. Comparison of accuracy, calibration (ECE), and discrimination (AUROC) across the three confidence estimation methods.*

There was no statistically significant difference between the three methods on raw benchmark accuracy. VCE had by far the best calibration error at only ~5.4%, with SC at ~9.4% and MSP at ~31.4%. VCE also

significantly outperformed in discrimination, with AUROC at ~0.76 compared to ~0.66 for SC and ~0.61 for MSP. According to the core performance metrics, VCE emerges as the clear overall winner.

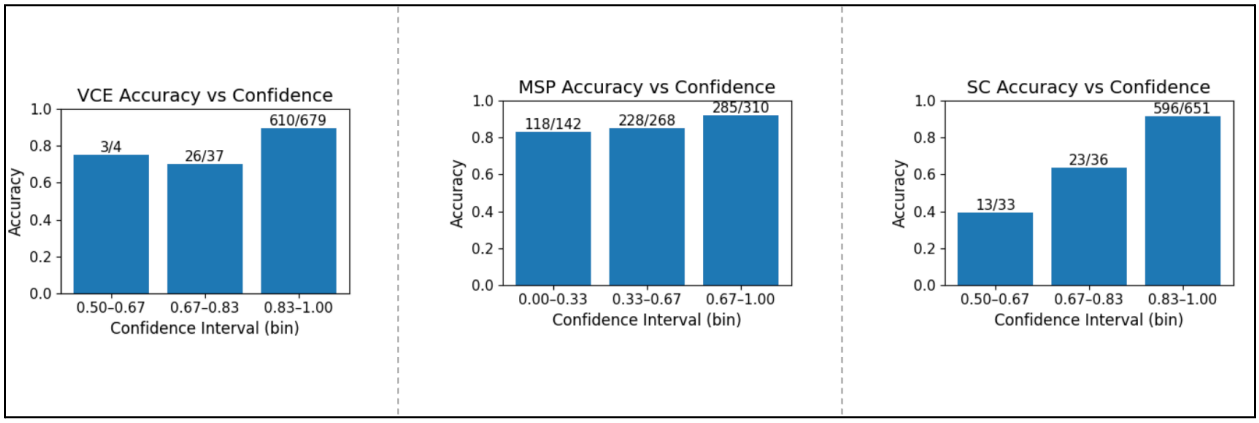

Figure 3. *Accuracy across three confidence intervals (bins) for VCE, MSP, and SC.*

A more in-depth picture emerges when specifically graphing accuracy vs confidence. Each graph in figure 3 splits the grade items according to three confidence intervals, which were obtained via equal division of possible confidence values for the corresponding method. In essence, the graphs compare the accuracy between low vs medium vs high confidence items for each method along with the number of items in each interval. Here we see that VCE does tend to have lower accuracy for low/medium confidence items, motivating the good ECE performance, but also that its confidence distribution is majorly skewed, with 610/679 items being high confidence. MSP has a smoother confidence distribution, but has minimal changes in accuracy compared to confidence. SC still has the skewed distribution (although milder than VCE), and presents remarkably clean-looking calibrations in accuracy corresponding to each confidence interval.

One final important result dimension can be obtained by looking at the confidence distribution of only incorrect items. Out of 81 incorrect for VCE, 69 (85%) were high confidence, 11 (14%) were medium confidence, and 1 (1%) were low confidence. For MSP (89 incorrect) it was 25 (28%) high, 40 (45%) medium, and 24 (27%) low. For SC (88 incorrect) it was 55 (62%) high, 13 (15%) medium, and 20 (23%) low.

4. Discussion

Properly contextualizing these results presents nuanced and informative takeaways. The core performance indicators (ECE, AUROC) suggest that VCE is by far the best confidence estimation method in this real-world benchmark context, but this conclusion comes with key caveats and each method has its pros and cons.

Verbalized Confidence Elicitation (VCE) has the most clear advantages, with its performance being the best on the metrics while being the easiest to implement across different models and tasks - it only requires some modifications and additions to the prompt. In addition, it produces human-interpretable confidence values due the opportunity for the LLM to present a rationale for each value, making it useful for diagnosing model behavior. For example, on an item “G” with confidence 0.7, the LLM explained that

"Confidence is lower for feature G due to ambiguity about whether mentioning a risk factor counts as a qualifier." However, VCE is extremely overconfident, with 85% of incorrect items still being high confidence - this overconfidence is a well-established flaw in prior literature (Hobelsberger et al., 2025).

In contrast, Maximum Sequence Probability (MSP) produces a smooth confidence distribution (likely due to the normalization), and it could be said that direct internal signals from an LLM are the most trustworthy. However, its implementation is flawed and limited especially in per-item granularity, and it lacks the ability to incorporate higher-level reasoning about ambiguity or task difficulty. As a result, MSP performs the worst across both confidence metrics. Additionally, it must be noted that its implementation is impossible for many frontier closed-source LLMs, as it requires token-level log probabilities to be accessible. Overall, this method has interesting research uses but lacks any capacity for usefulness in real-world applications.

Sample Consistency (SC) provides the advantages of decent performance and stable outputs. In addition, it is less overconfident than VCE, and its implementation is intuitive and simple. With generating multiple samples already being useful for just output robustness, getting a somewhat useful confidence metric could be framed as a nice bonus. However, its major drawback is efficiency. Because it requires multiple model generations per input, SC is extremely slow and computationally expensive, making it difficult to scale in real-world systems or cost-sensitive environments.

Overall, these trade-offs highlight the importance of choosing the right method for deployment. However, it must also be acknowledged that none of the three methods can currently be framed as being directly applicable as a task routing mechanism in real workflows. Note that the better performing methods (VCE and SC) were excessively overconfident in their confidence values. This essentially means that using either of these methods as a routing mechanism would result in the majority of incorrect items slipping right through as "high confidence" without human intervention. Also note that for methods other than VCE, sending "low confidence" items to human review would result in human experts spending time reviewing items of which a vast majority were still correct.

5. Conclusion

This study explored how different uncertainty estimation methods would perform in a realistic task-routing setting, where LLM confidence must be measured during its grading of student answers in order to decide when human faculty intervention is needed. The results show that Verbalized Confidence Elicitation (VCE) provides the best performing and most practicable method amongst those tested according to our metrics, while also having the best interpretability. However, our results also showed that none of the methods are ready for real-world implementation.

This highlights a key issue in the field of LLMs: current LLM uncertainty estimation methods are poor and insufficient. In other words, there does not yet exist a way for humans to know whether they can "trust" an LLM's output. For many real use-cases, this is a fatal flaw - if there is no good way to know which outputs require human review, then all outputs must be reviewed, which is often not feasible. Thus, this study highlights what could be framed as one of the main blockers to widespread LLM adoption and integration in the real world. Uncertainty/confidence estimation is not just a niche research topic, but a necessary design component in human-AI systems.

While this study focuses on a specific dataset and task, it opens the door for further research into more robust and generalizable uncertainty estimation techniques. Future work may explore newer or more advanced estimation methods (with many more currently under active research and development), evaluating methods on other LLMs of different capabilities or architectures, and of course evaluation on more datasets and tasks. In particular, we have open-sourced our code and the de-identified dataset/benchmark on a public GitHub repository, such that anyone interested can expand on our work. Ultimately, advancing LLM uncertainty estimation will be essential for building AI-integrated systems that are not only accurate, but also transparent, accountable, and safe to deploy at scale.

References

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